

2 Inheritance of wing shape and eye colour in the fruit fly, *Drosophila melanogaster*, is controlled by two genes.

- Gene **N/n** controls wing shape. Allele **N** for wrinkled wings is dominant to allele **n** for normal wings.
- Gene **E/e** controls eye colour. Allele **E** for rosy eyes is dominant to allele **e** for red eyes.

A biologist predicted that, if the genes are on **different** chromosomes, the ratio of the phenotypes of the F₂ generation would be 9:3:3:1.

The biologist carried out a breeding experiment.

- Homozygous dominant fruit flies with wrinkled wings and rosy eyes were crossed with homozygous recessive fruit flies with normal wings and red eyes.
- All the F₁ fruit flies had wrinkled wings and rosy eyes.
- The F₁ fruit flies were crossed with each other.

Table 2.1 shows the results for the F₂ generation.

Table 2.1

F2 phenotype	frequency
wrinkled wings rosy eyes	44
wrinkled wings red eyes	2
normal wings rosy eyes	2
normal wings red eyes	16
total	64

(a) The chi-squared test (χ^2 test) was used to analyse the data in Table 2.1.

(i) State **one** reason why the chi-squared test (χ^2 test) was used.

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..... [1]

(ii) State the null hypothesis that the biologist would use for this test.

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..... [1]

(iii) Complete Table 2.2 **and** calculate the value of χ^2 for the results of the F2 generation.

The equation for the calculation of χ^2 is:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

O = observed result

E = expected result

Σ = sum of

Table 2.2

offspring phenotype	O	E	$\frac{(O-E)^2}{E}$
wrinkled wings rosy eyes	44		
wrinkled wings red eyes	2		
normal wings rosy eyes	2		
normal wings red eyes	16		
$\chi^2 =$			

[3]

Table 2.3 shows some critical values of χ^2 at different probability levels.

Table 2.3

degrees of freedom	probability (p)			
	0.10	0.05	0.01	0.001
1	2.71	3.84	6.64	10.83
2	4.61	5.99	9.21	13.82
3	6.25	7.82	11.34	16.27
4	7.78	9.49	13.28	18.46

(iv) State why the student used the critical value of **7.82** in this investigation.

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..... [1]

(v) Use your calculated value of χ^2 to:

- explain whether the null hypothesis should be accepted or rejected

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- suggest a conclusion the biologist could make about the inheritance of the genes controlling wing shape and eye colour in fruit flies.

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..... [3]

[Total: 9]